

level 1

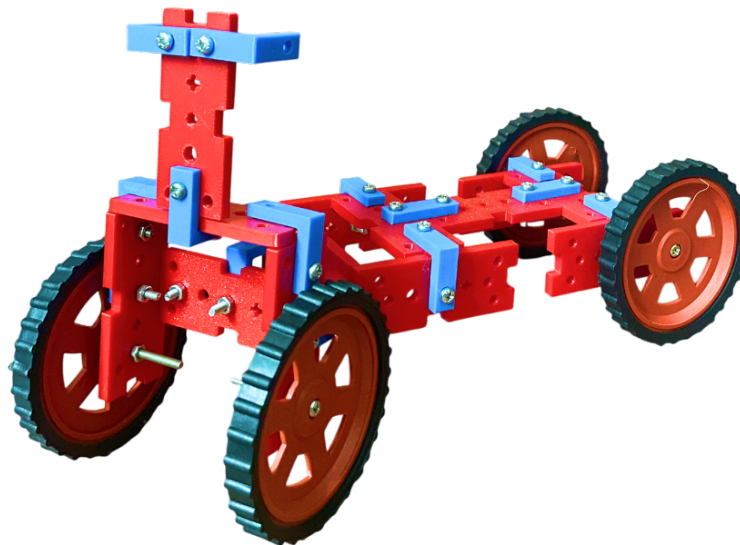
oh.

rickshaw

a fun machine with a cart that shows how wheels make transport easy! I have 4 wheels and a cart to carry the  weight of passengers. When you push me, my wheels reduce friction, helping me move smoothly with less effort!

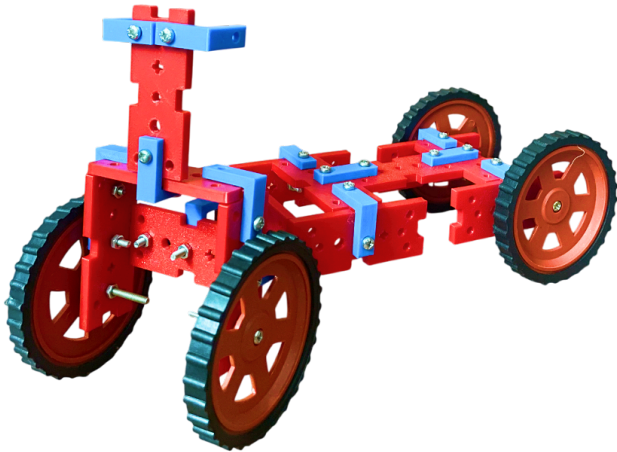
no. of sessions: 6

simple machine: wheel

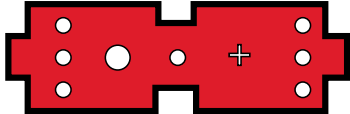


level 1

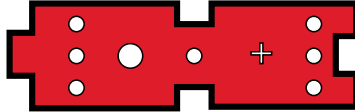
oh.



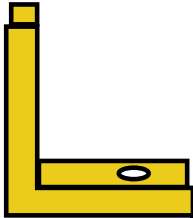
materials needed



m-m beam - x1



m-f beam - x3



L clamp - x4

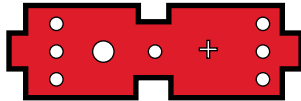


small screw - x9

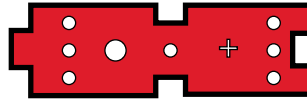


nut - x9

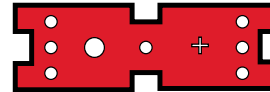
materials needed



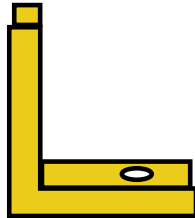
m-m beam - x1



m-f beam - x2



f-f beam - x1



L clamp - x4

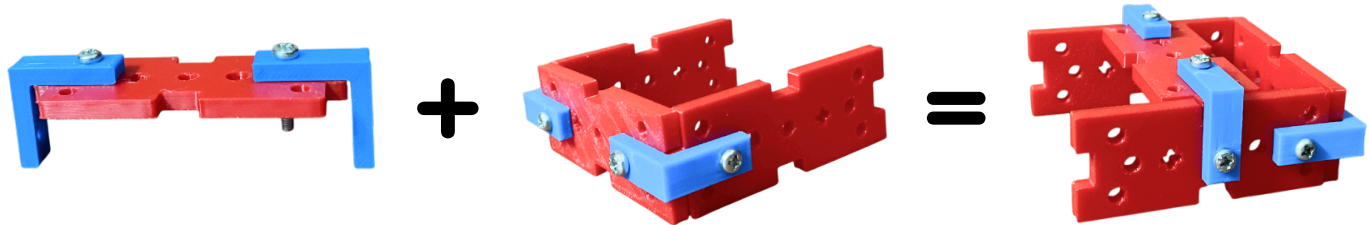


small screw - x9



nut - x9

step 1 : assemble the front body



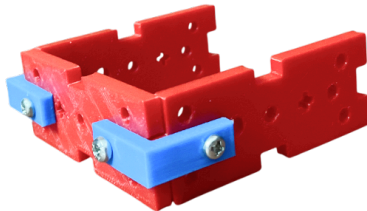
Attach **2x L clamps** the ends of the M-M beam.

Lets join **3 x M-F beams** to form a U shape.

Use **2 L clamps** to hold them together.

Attach the both the pieces together as shown.

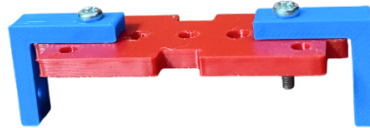
step 1 : assemble the front body



Lets join **3 x M-F beams** to form a U shape.

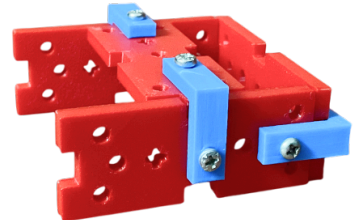
Use **2 L clamps** to hold them together.

+



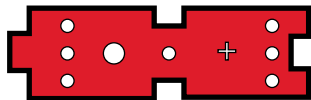
Attach **2x L clamps** the ends of a M-M beam.

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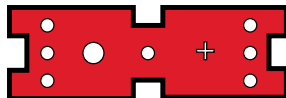


Attach the both the pieces together as shown.

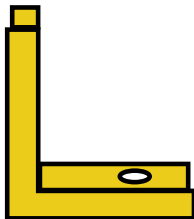
materials needed



m-f beam - x2



f-f beam - x1



L clamp - x3

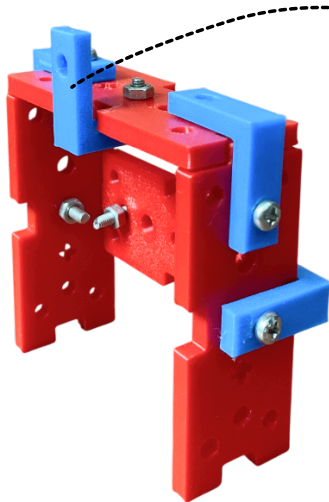


small screw - x3

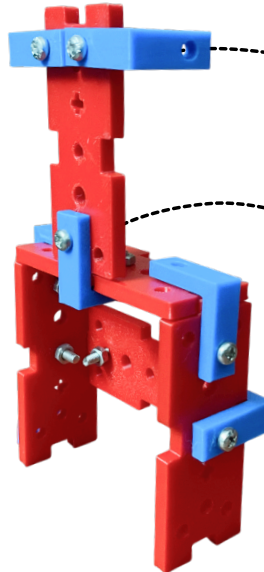


nut - x3

step 2: assemble the rickshaw handle



Attach a **L clamp** to the U shape structure as shown.



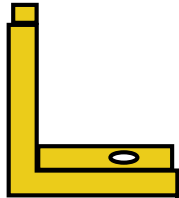
Add **2 more L clamps** for the handles on the other end of the M-F beam.

Join a **M-F beam** to the L clamp as shown.

materials needed



wheel - x2



L clamp - x1



long screw - x2



small screw - x1

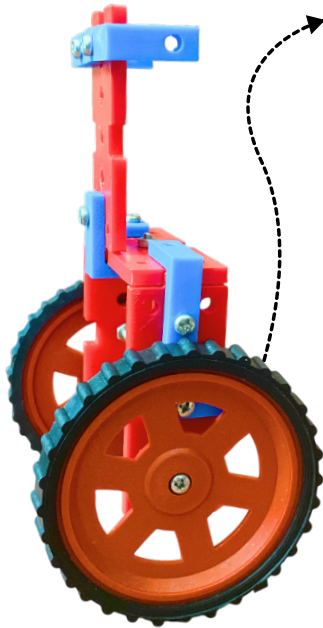


nut - x3



**Do not tighten the screw too much,
keep the screw slightly loose for
the wheel to move freely.**

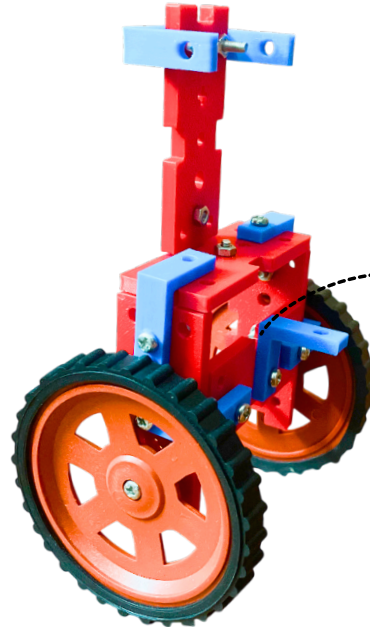
step 3: attach wheels to the front body



Attach the **2 wheels** using long screws to both sides of the rickshaw.

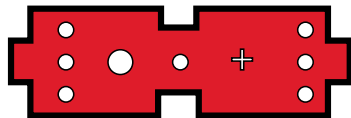
Use 2 nuts to tighten each screw

With this step the front part of the Rickshaw is complete.



Add a **L clamp** to back of the beam as shown

materials needed



m-m beam - x3



I clamp - x2

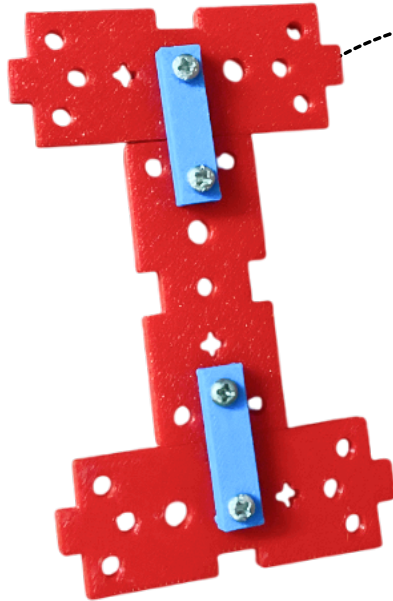


small screw - x4



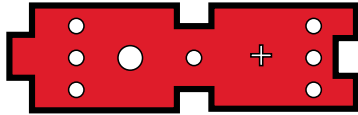
nut - x4

step 4: assemble the cart for the rickshaw

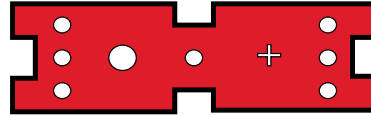


Join 3 x M-M beams
together using I clamps
to form an I shape.

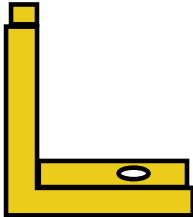
materials needed



m-f beam - x2



f-f beam - x1



L clamp - x2

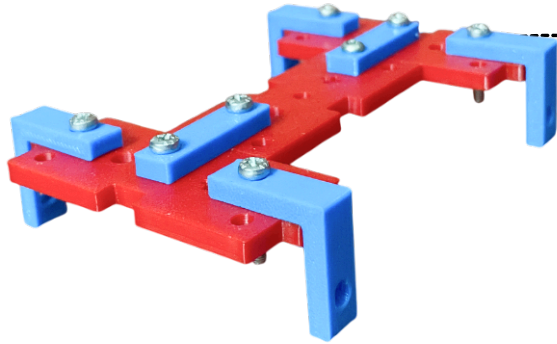


small screw - x4

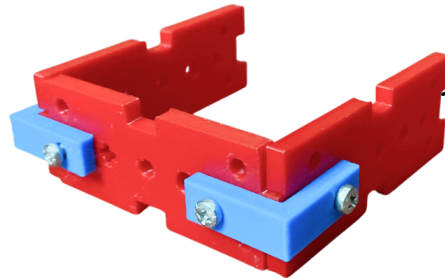


nut - x4

step 5: attach the supports to the cart



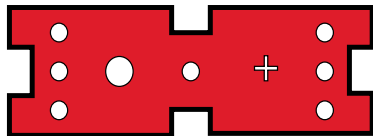
Fix **4 x L clamps** on the both ends of the Beam as shown. These supports will be used to hold the cart together.



Lets join **2 x M-F** and **1 F-F beam** to form a U shape.

Use **2 L clamps** to hold them together.

materials needed



f-f beam - x2

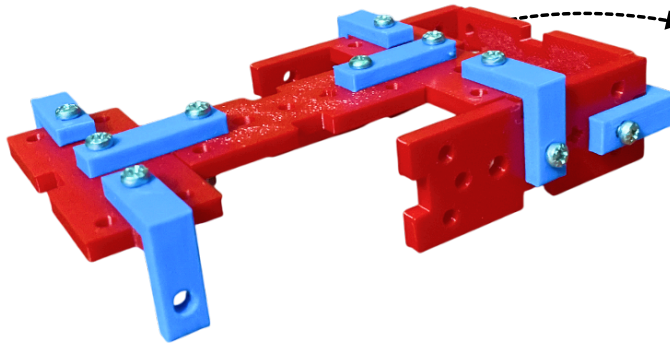


small screw - x4

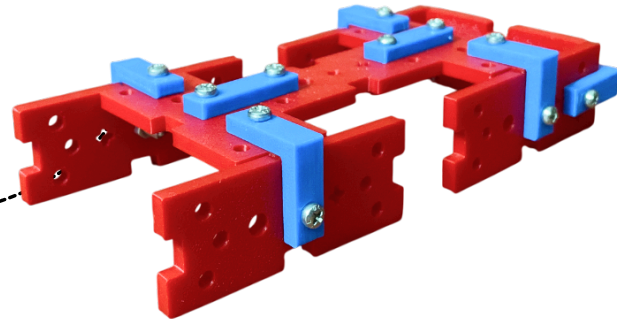


nut - x4

step 6: complete the cart assembly



Join the U shape structure to the cart as shown.



Attach 2 F-F beams to the remaining sides of the L clamp as shown

materials needed



wheel - x2



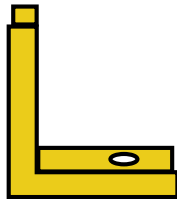
long screw - x2



nut - x3

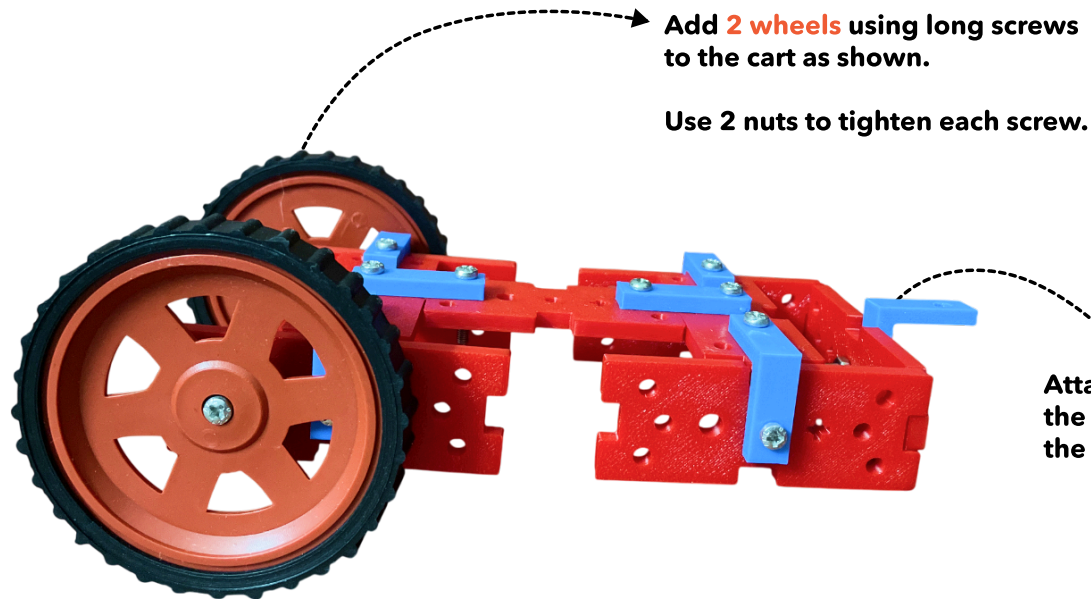


small screw - x1



L clamp - x1

step 7: assemble the wheels to the cart



Add **2 wheels** using long screws to the cart as shown.

Use 2 nuts to tighten each screw.

Attach a **L clamp** to the front beam of the cart as shown

materials needed

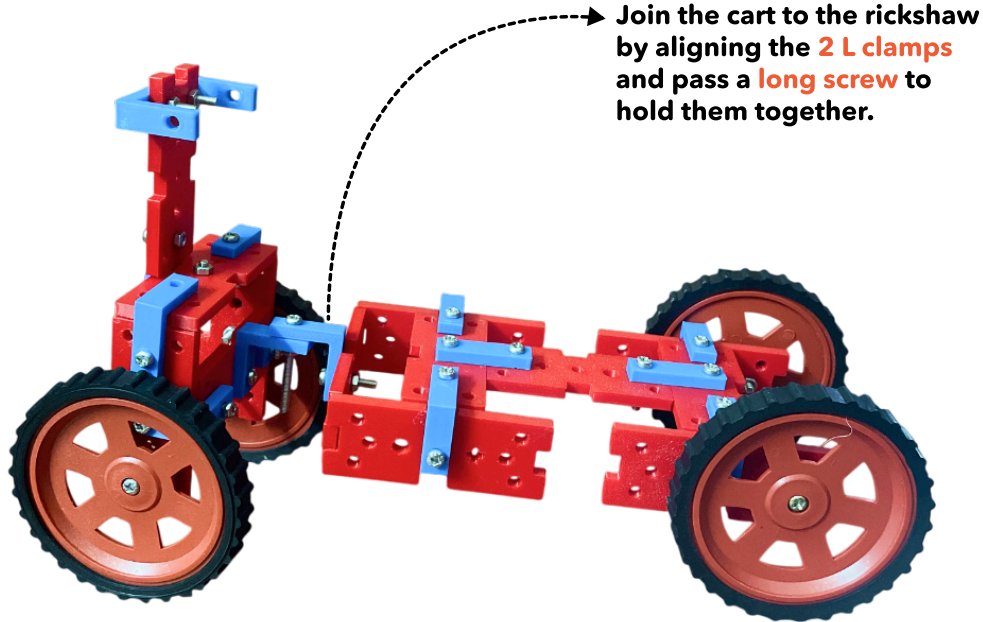


long screw - x1



nut - x2

step 8: set it up and test



TEST IT OUT!

Place some  into the cart and take your rickshaw for a spin!